HACKARENA 2.0

TEAMNAME: HALLUCINATING CRUSADERS

COLLEGE NAME: PES UNIVERSITY

PROBLEM STATEMENT

- Small businesses often face hurdles while dealing with security like facing thousands of critical alerts generated across tools like Wazuh, Okta, and AWS, but lack the massive budgets required for enterprise SIEM platforms and the dedicated teams needed to maintain complex open-source pipelines involved.

- Often solo administrators deal with massive amount of alerts which often sidelines a lot of security events leaving the businesses vulnerable to breaches.

- This is caused by the lack of zero-config, lightweight cross-vendor reasoning solutions tailored for resource-constrained SMBs, and the critical vulnerability of prompt injection, where an attacker can easily hijack a standard AI security script by embedding malicious instructions inside raw server logs leading to false negatives

OUR SOLUTION

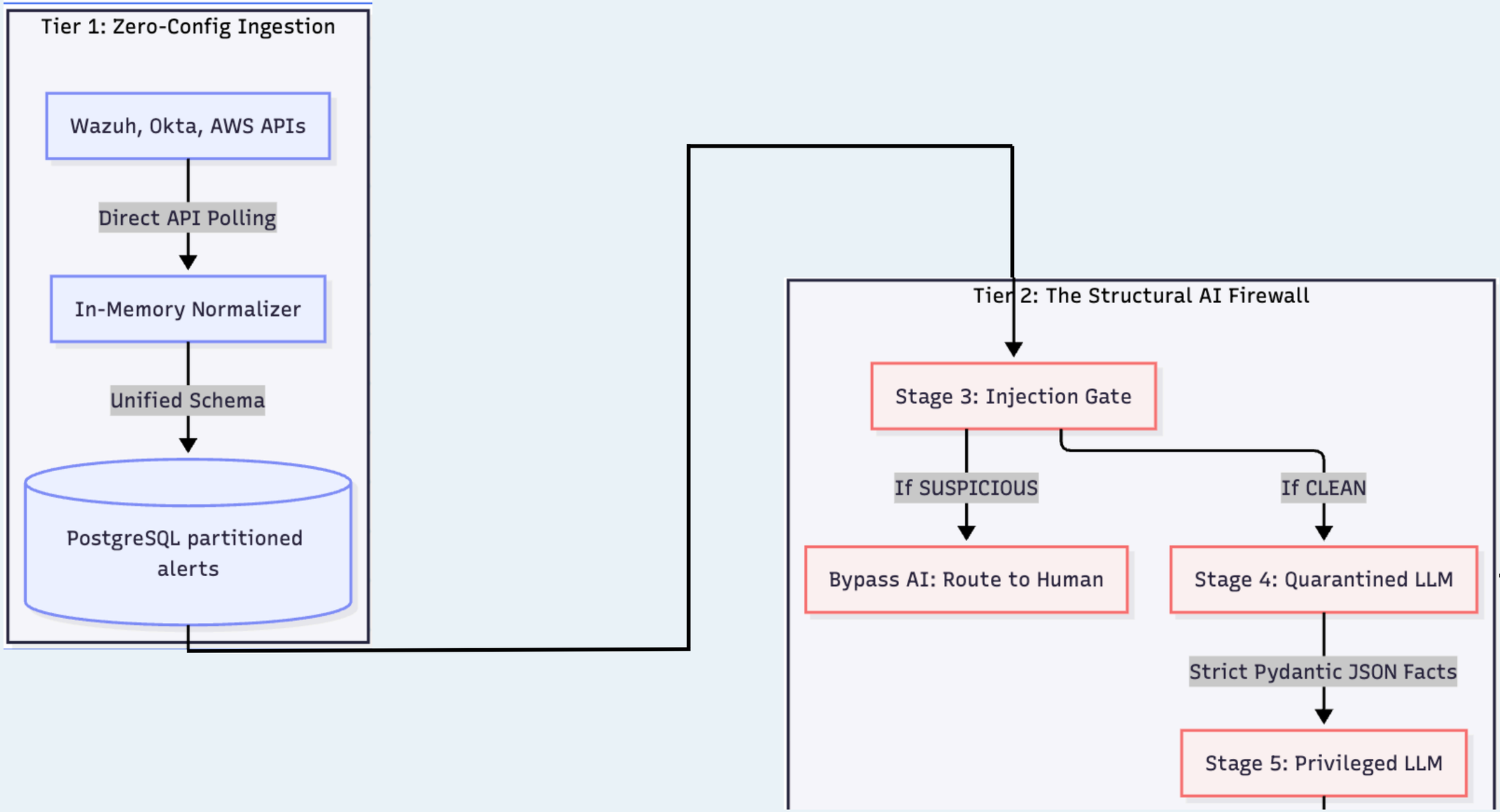
An autonomous, AI-powered Security Operations Center (SOC) analyst system that monitors a Windows host in real-time through Wazuh SIEM, processes security alerts through a stateful multi-agent LangGraph pipeline, enriches them with multi-vendor threat intelligence, and presents findings on a premium real-time dashboard with Human in-the-Loop (HITL) active containment capabilities.

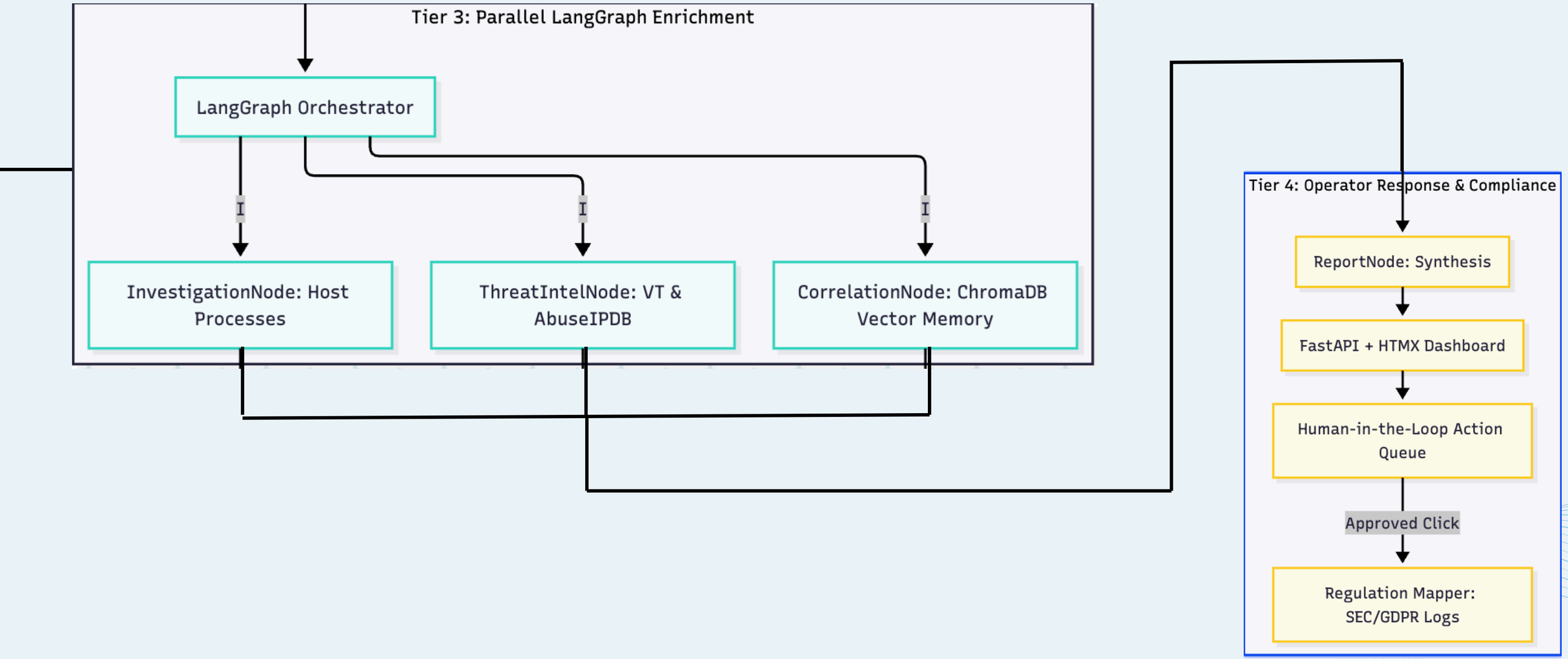
- The system bypasses heavy enterprise data lakes and expensive software contracts by using direct API polling to instantly pull, normalize, and correlate alerts across an SMB's existing, fragmented tools (like Wazuh, Okta, and AWS) in temporary memory.
- It introduces a dual-LLM architecture featuring a non-AI pre-input gate and an isolated data extraction model that ensures that raw, untrusted hacker logs never reach the primary reasoning AI, making it immune to prompt manipulation.
- Built to be lightweight and accessible, the entire application can be deployed via simple API keys, delivering a real-time, dark-themed dashboard where a solo IT admin can review AI-generated markdown reports and securely approve automated fixes with a single click.

PROBLEMS SOLVED

Existing solutions	Our solution
Focussed on large enterprieses, require a dedicated DevOps team to configure and maintain massive data pipelines like Kafka as data must be accumulated and managed from a central database	Normalizes alerts directly in application memory via simple API polling, cutting deployment time down massively for a solo administrator.
Rely on weak text-based prompting to ignore hacker tricks hidden inside logs, which routinely fails.	Implements a strict, physical dual-LLM boundary where raw, untrusted alert data never reaches the primary reasoning brain, preventing prompt injection attacks.
Fail to document why an AI chose to take a destructive action.	Introduces a regulatory regulation mapper that automatically links every automated fix directly to its specific evidence log IDs and pairs it with statutory disclosure timelines (lEx: SEC 4-day mandate)
Processes alerts sequentially based on simple "If-This-Then-That" automation	Uses LangGraph to fan out data collection across specialized agents concurrently. This compresses a deep, cross-vendor security investigation down from minutes to a 30-second live dashboard update

ARCHITECTURE




Tier 3: Parallel LangGraph Enrichment
LangGraph Orchestrator
InvestigationNode: Host Processes
ThreatIntelNode: VT & AbuseIPDB
CorrelationNode: ChromaDB Vector Memory
Tier 4: Operator Response & Compliance
ReportNode: Synthesis
FastAPI + HTMX Dashboard
Human-in-the-Loop Action Queue
Approved Click
Regulation Mapper: SEC/GDPR Logs

ANALYSIS

The system ingests raw security alerts from Wazuh SIEM via direct API polling and normalizes them in-memory, eliminating the need for heavy data pipelines. A dual-LLM architecture ensures untrusted log data is sanitized through a non-AI pre-input gate before reaching the reasoning model, preventing prompt injection attacks. The LangGraph multi-agent pipeline then fans out concurrent investigations across vendors – correlating alerts from Wazuh, Okta, and AWS – compressing deep cross-vendor analysis from minutes down to approximately 30 seconds per alert cycle.

RESULT

- Dual-LLM architecture blocked all prompt injection attempts – zero false negatives from malicious logs.
- Cross-vendor alert correlation across Wazuh, Okta, and AWS completed in under 30 seconds per cycle.
- Single-click Human-in-the-Loop containment makes enterprise-grade security accessible to solo admins.

CONCLUSION

Our AI-powered SOC analyst proves that small businesses no longer need enterprise budgets or dedicated security teams to maintain robust threat detection. The dual-LLM architecture and concurrent multi-agent pipeline deliver fast, reliable, and prompt-injection-proof security operations through a simple zero-config deployment. With built-in regulatory compliance mapping, every automated action remains audit-ready – turning a solo IT admin into a fully equipped security operations center.

THANK YOU